Milestone 3: How many buggy classes could have been prevented by having zero smells?

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Project Selection

- Each student must work on **one project as** identified via this algorithm:

1. Take the first letter of your last name (e.g., F for Falessi)
2. Take the number of the letter (e.g., F = 6)
3. X = number mod 6 (e.g. X = 0)

```
switch (X) {
    case 0: project == AVRO;
    case 1: project == OPENJPA;
    case 2: project == STORM;
    case 3: project == ZOOKEEPER;
    case 4: project == SYNCOPE;
    case 5: project == TAJO;
    }
```

Your Project – What-if?

1. Create A (Milestone 1)
2. Compare the accuracy of three classifiers on A (Milestone 2)
3. Choose the best classifier, aka, BClassifier.
5. Create:
    1. B+: Portion of A with NSmells >0
    2. C portion of A with NSmells =0.
    3. B: B+ manipulated with feature NSmells set to 0
6. Train BClassifier on A, aka BClassifierA
7. Predict A, B, B+, C and create a Table like in slide 14

Results: What if analysis

- Is BClassifier accurate?

- How many buggy classes could have been prevented by having zero smells?
  - In total
  - In proportion
  - Out of the preventable ones